FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 3.2: PROBABILITY I	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent 8 lessons investigating probability, independent events, and the mathematics behind chance. Now it is time to bring everything together and answer the driving question using evidence, mathematical reasoning, and real-world connections.

READ THE DRIVING QUESTION CAREFULLY:
"If past outcomes of a chance event — like a coin toss or a football match — do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?"

YOUR TASK: Write a complete, well-reasoned explanation that answers every part of the driving question. Your response must:

- Use correct mathematical vocabulary (probability, equally likely events, mutually exclusive events, sample space, independent events, the gambler's fallacy, theoretical vs. experimental probability).
- Include at least TWO mathematical calculations or expressions to support your reasoning.
- Reference the anchoring phenomenon (Gor Mahia FC's winning streak and betting culture in Nairobi).
- Connect probability mathematics to real-life decision-making and the dangers of gambling.
- Be written in full sentences and organised into clear sections using the headings provided below.

Work through each section in order. The exemplar response beneath each prompt shows you the level of thinking expected — use it as a guide, NOT something to copy.

Section 1: What Is Probability and How Do We Measure It?

In your own words, explain what probability is and how it is measured. Define the probability scale from 0 to 1 (or 0% to 100%) and explain what	Probability is a measure of how likely it is that a particular event will happen. We express probability as a number between 0 and 1, where 0 means the event is impossible and 1 means the event is certain. For example, the probability that the sun will rise tomorrow over Nairobi is 1 — it is certain. The probability that a githeri grain chosen at random from a pot of 10 beans and 10 maize is a bean is $10/20 = 1/2 = 0.5 = 50\%$. This tells us there is an equal chance of picking either a bean or a maize grain. Events with a probability close to 0 are very unlikely, events close to 0.5 are roughly even, and events close to 1 are very likely.

values at each end mean. Use at least ONE example from everyday Kenyan life — this could involve a coin, a die, a football match, or a common activity like drawing a card or picking a bean from a bag. Make sure you write the probability as a fraction, decimal, and percentage.	Understanding this scale is the first step in reading any probability situation mathematically, rather than relying on gut feeling or superstition.

Section 2: Equally Likely Events, Sample Space, and Mutually Exclusive Events	
Explain what it means for events to be equally likely, and define what a sample space is using a specific example. Then explain what mutually exclusive events are and give ONE example from the Gor Mahia FC phenomenon — for instance, the possible outcomes of a single football match. Write out the sample space for your chosen example and calculate the probability of at least one of the mutually exclusive outcomes.	Equally likely events are outcomes that each have the same chance of occurring. For example, when a fair coin is tossed, the two possible outcomes — heads (H) and tails (T) — are equally likely. The sample space is the complete list of all possible outcomes. For a single coin toss, the sample space is S = {H, T}, so there are 2 equally likely outcomes. When we consider the result of a Gor Mahia FC match, the sample space is S = {Win, Draw, Loss}. A Win and a Loss are mutually exclusive events — they cannot both happen in the same match at the same time. If we assume each outcome is equally likely, then $P(\text{Win}) = 1/3 \approx 0.33 = 33\%$. Mutually exclusive events are important because knowing one outcome happened immediately tells us the other outcomes in that set did not. However, just because Gor Mahia won their last match does NOT mean a loss is now more likely in the next one — that would be a mathematical error called the gambler's fallacy.

Section 3: Independent Events and the Gambler's Fallacy	
This is the heart of the driving question. Explain clearly what an independent event is in mathematical terms. Then describe the gambler's fallacy — the mistaken belief that past results change future probabilities. Use the Gor Mahia winning streak AND a coin-toss example to show mathematically why the gambler's fallacy is wrong. You must include a calculation	Two events are independent if the outcome of one has absolutely no effect on the probability of the other. A coin toss is a perfect example: no matter how many times you toss the coin, each new toss is a fresh start with $P(\text{Heads}) = 1/2$. Suppose you toss a fair coin and get Heads five times in a row. Many people feel that Tails is 'overdue' and must be more likely on the 6th toss. This is the gambler's fallacy — a false belief. Mathematically, the probability of Heads on the 6th toss is still exactly $P(\text{Heads}) = 1/2 = 0.5 = 50\%$. The coin has no memory. The same principle applies to Gor Mahia's matches. Each football match is an independent event. If Gor Mahia won 8 matches in a row, commentators who say the team is 'due for a loss' are committing the gambler's fallacy. The probability of the result in match 9 is not affected by the previous 8 results. Sports betting companies and advertisers exploit this fallacy deliberately — they encourage fans to believe patterns in past results reveal future certainties, when in reality each match resets. Recognising independence mathematically protects you from being misled.

showing the probability of getting heads on a 6th toss after already getting heads 5 times in a row.	

Section 4: Theoretical vs. Experimental Probability — What Patterns Can We Trust?	
Explain the difference between theoretical probability and experimental probability. Describe what happens when you repeat a chance experiment many times (refer to the Law of Large Numbers in simple terms). Use an example from your classwork — such as tossing a coin 10 times vs. 100 times, or recording the outcomes of simulated Gor Mahia matches — to explain why experimental probability gets closer to theoretical probability over many trials, but this does NOT mean any single outcome is 'due.' Finish by explaining what mathematics CAN and CANNOT tell us about predicting individual outcomes.	Theoretical probability is what we calculate using mathematics based on equally likely outcomes. For a fair die, the theoretical probability of rolling a 3 is $P(3) = 1/6 \approx 0.167 = 16.7\%$. Experimental probability is what we actually observe when we carry out a real experiment. If you toss a coin 10 times and get 7 Heads, the experimental probability of Heads is $7/10 = 0.7 = 70\%$ — which is far from the theoretical 50%. However, if you toss the coin 1,000 times, the experimental probability will get much closer to 50%. This is sometimes called the Law of Large Numbers: over many trials, experimental results approach theoretical probability. This is a statement about long-run patterns, NOT individual events. It does NOT mean that after 7 Heads in a row, Tails is 'catching up.' Each toss is still 50/50. What mathematics CAN tell us is what to expect on average over many events. What it CANNOT do is predict the result of any single toss or any single football match with certainty. Anyone — including a betting company — who claims they can reliably predict one match outcome is misusing probability.

Section 5: Answering the Driving Question — Probability as Protection	
Now write your complete answer to the driving question. Bring together everything from Sections 1–4. Your answer must: (1) explain how mathematics helps us make predictions about chance events; (2) explain clearly why past outcomes do NOT change future probabilities for independent events; (3) explain how the gambler's fallacy makes gambling dangerous,	Mathematics gives us a powerful tool for understanding chance. By calculating probabilities, identifying sample spaces, and recognising independent events, we can make informed predictions about what is likely to happen over time. For example, we know that if a fair coin is tossed repeatedly, roughly half the outcomes will be Heads — not because any toss is 'owed,' but because each independent toss carries a 50% probability. The same principle applies to football matches: each game Gor Mahia FC plays is a new, independent event with its own probability of a Win, Draw, or Loss. The 2023 Premier League winning streak did not make a future loss more likely — mathematics tells us this clearly. The danger of gambling lies precisely in exploiting the gambler's fallacy. Betting advertisements on Nairobi matatus and social media tell young people that they can 'read the pattern' and predict outcomes. But because each match is an independent event, no pattern from the past changes the mathematics of the future. Betting companies profit because most players do not understand probability — they believe luck is 'due' and keep betting to recover losses, a cycle that leads to financial harm. Understanding probability protects me as a young Kenyan by helping me evaluate risk honestly. When I see a betting advert claiming I can 'beat the odds,' I now know that no amount of past data removes the fundamental uncertainty of an independent chance event. Mathematics does not just help me pass exams — it helps me make safer, smarter decisions in real life.

using the Nairobi betting culture context; and (4) reflect personally on why this mathematical understanding is important for your own decision-making as a young person in Kenya today.	

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Probability Concepts	Accurately defines probability, the 0–1 scale, equally likely events, sample space, and mutually exclusive events. Uses correct mathematical vocabulary consistently throughout all sections with no errors.	Correctly defines most key probability concepts with minor omissions. Uses mathematical vocabulary appropriately in most sections with only 1–2 small errors.	Shows partial understanding of probability concepts. Definitions are incomplete or contain errors. Mathematical vocabulary is used inconsistently or incorrectly in multiple places.
Mathematical Accuracy and Use of Calculations	Includes at least TWO accurate, clearly shown calculations (fractions, decimals, and percentages used correctly). All probability values are within the range [0, 1]. Working is clearly shown and logically supports the written explanation.	Includes at least TWO calculations that are mostly correct. Probability values are within range. Minor arithmetic or notation errors present but do not undermine the overall reasoning.	Fewer than two calculations are attempted, OR calculations contain significant errors. Probability values may fall outside the range [0, 1]. Working is unclear or missing.
Explanation of Independent Events and the Gambler's Fallacy	Clearly and accurately explains what independence means mathematically. Powerfully debunks the gambler's fallacy using both the coin-toss example and the Gor Mahia FC context, with supporting calculation. Explanation leaves no ambiguity about why past outcomes do not affect future probabilities.	Explains independence and the gambler's fallacy correctly but with less mathematical precision or depth. Both examples (coin toss and Gor Mahia) are referenced but one may be underdeveloped.	Attempts to explain independence or the gambler's fallacy but contains misconceptions or confuses the two concepts. May only reference one example or omit a calculation entirely.
Connection to Real-Life Context and Gambling Awareness	Explicitly connects probability concepts to the Nairobi betting culture and Gor Mahia FC phenomenon throughout the explanation. Clearly articulates how the gambler's fallacy enables gambling harm. Personal reflection in Section 5 is specific, sincere, and mathematically grounded.	References the Gor Mahia FC phenomenon and betting culture in at least two sections. Gambling harm is acknowledged. Personal reflection is present but may lack mathematical specificity.	Makes only a surface-level or single reference to the Kenyan context. The link between the gambler's fallacy and gambling harm is vague or missing. Personal reflection is absent or generic.
Organisation, Clarity, and Use of Mathematical Language	All five sections are complete and clearly address their prompts. Writing is well-organised, easy to follow, and uses	All five sections are present and mostly address their prompts. Writing is generally clear. Mathematical language is	One or more sections are missing or do not address the prompt. Writing is sometimes unclear or disorganised.

	precise mathematical language throughout. Ideas flow logically from one section to the next, building toward a strong final answer in Section 5.	used correctly in most places. The final answer in Section 5 brings most ideas together.	Mathematical language is imprecise or used incorrectly. Section 5 does not successfully synthesise earlier sections.